Name:

STN:

In the Name of God

Deep Learning

Computer Assignment 3

• Identifying Different Flowers Using Convolutional Neural Networks

The goal of this assignment is to design a model using convolutional neural networks. The dataset provided to you contains 5 different classes of flowers.

In this assignment, you must design a network that can identify which of these classes an input image belongs to. To do this, we will design a convolutional network. First, choose the architecture of your network as you see fit, and try to achieve the best possible accuracy by modifying it. Report your results.

1) Choose appropriate hyperparameters for the network and explain how you obtained these hyperparameters.

2) Make appropriate changes to the network to prevent overfitting.

3) Train the network and report its accuracy on the training data.

4) Use Recall and Precision metrics to evaluate the network's performance on the training data and evaluation data.

5) Plot the error curve during the training process.

6) Use several optimizer functions introduced during the course to train the network, and report the changes obtained.

7) Prepare two images of your choice, provide them as inputs to the neural network you designed, and show the network output.

Now that you have designed and trained your model, answer the following questions:

1) What is the highest accuracy you can achieve using the model?

2) What are the best optimizer and learning rate for the network? Provide results for several different optimizers and learning rates, and compare them with one another.

Additional Notes:

1) The starter code has been provided to you. If you have difficulty running it on your system, use the Google Colab environment.

2) For the Persian explanations and report, there is no need for a separate PDF file; write the Persian explanations in the same .ipynb file.

3) Use deep learning libraries such as TensorFlow and Keras to complete this assignment.

4) To increase execution speed, you can select a GPU as the processor type.

5) This is an individual assignment and copying is not permitted. However, you may consult others regarding experience with writing code and analyzing and comparing results.

Good luck!